Learning Objectives

On completion of this chapter, the students are expected to know

- the concept of Binomial Theorem, to compute binomial coefficients and their applications
- the concepts of sequences and series
- how to compute arithmetic, geometric and harmonic means
- how to find the sum of finite and infinite series of real numbers
- how to add series using telescopic summation
- how to apply binomial, exponential and logarithmic series

5.2 Binomial Theorem

The prefix **bi** in the words bicycle, binocular, binary and in many more words means **two**. The word ***binomial*** stands for expressions having two terms. For examples $(1+x),\ (x+y),\ (x^2+xy)$ and $(2a+3b)$ are some binomial expressions.

5.2.1 Binomial Coefficients

In Chapter 4 we have learnt and used the symbol ${}^{n}\mathrm{C}_{r}$ which is defined as

$$ {}^{n}\mathrm{C}_{r} = \frac{n(n-1)(n-2)\dots(n-(r-1))}{r(r-1)(r-2)\dots 1} = \frac{n!}{(n-r)!\,r!}. $$

Since ${}^{n}\mathrm{C}_{r}$ occurs as the coefficients of x^r in $(1+x)^n$ $n \in \mathbb{N}$ and as the coefficients of $a^r b^{n-r}$ in $(a+b)^n$, they are called binomial coefficients. Though the values of ${}^{n}\mathrm{C}_{r}$ can be computed by formula, there is an interesting simple way to find ${}^{n}\mathrm{C}_{r}$ without doing cumbersome multiplications.

Pascal Triangle

The Pascal triangle is an arrangement of the values of ${}^{n}\mathrm{C}_{r}$ in a triangular form. The $(k+1)^{st}$ row consists values of

$$ {}^{k}\mathrm{C}_{0}\,,\ {}^{k}\mathrm{C}_{1}\,,\ {}^{k}\mathrm{C}_{2}\,,\ {}^{k}\mathrm{C}_{3}\,,\dots,\ {}^{k}\mathrm{C}_{k}\,. $$

In fact, the Pascal triangle is

$$
\begin{array}{ccc}
{}^{0}\mathrm{C}_{0} & & 1 \\
{}^{1}\mathrm{C}_{0}\quad {}^{1}\mathrm{C}_{1} & & 1\quad 1 \\
{}^{2}\mathrm{C}_{0}\quad {}^{2}\mathrm{C}_{1}\quad {}^{2}\mathrm{C}_{2} & & 1\quad 2\quad 1 \\
{}^{3}\mathrm{C}_{0}\quad {}^{3}\mathrm{C}_{1}\quad {}^{3}\mathrm{C}_{2}\quad {}^{3}\mathrm{C}_{3} & \Leftrightarrow & 1\quad 3\quad 3\quad 1 \\
{}^{4}\mathrm{C}_{0}\quad {}^{4}\mathrm{C}_{1}\quad {}^{4}\mathrm{C}_{2}\quad {}^{4}\mathrm{C}_{3}\quad {}^{4}\mathrm{C}_{4} & & 1\quad 4\quad 6\quad 4\quad 1 \\
\dots\quad \dots\quad \dots\quad \dots & & \dots\quad \dots\quad \dots\quad \dots \\
\dots\quad \dots\quad \dots\quad \dots & & \dots\quad \dots\quad \dots\quad \dots
\end{array}
$$